Context-Aware Local–Global Semantic Alignment for Remote Sensing Image–Text Retrieval

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***Abstract*— Remote sensing image–text retrieval (RSITR) is a cross-modal task that integrates visual and textual information, attracting significant attention in remote sensing research. Remote sensing images typically contain complex scenes with abundant details, presenting significant challenges for accurate semantic alignment between images and texts. Despite advances in the field, achieving precise alignment in such intricate contexts remains a major hurdle. To address this challenge, this article introduces a novel context-aware local–global semantic alignment (CLGSA) method. The proposed method consists of two key modules: the local key feature alignment (LKFA) module and the cross-sample global semantic alignment (CGSA) module. The LKFA module incorporates a local image masking and reconstruction task to improve the alignment between image and text features. Specifically, this module masks certain regions of the image and uses text context information to guide the reconstruction of the masked areas, enhancing the alignment of local semantics and ensuring more accurate retrieval of region-specific content. The CGSA module employs a hard sample triplet loss to improve global semantic consistency. By prioritizing difficult samples during training, this module refines feature space distributions, helping the model better capture global semantics across the entire image–text pair. A series of extensive experiments demonstrates the effectiveness of the proposed method. The method achieves an mR score of 32.07% on the RSICD dataset and 46.63% on the RSITMD dataset, outperforming baseline methods and confirming the robustness and accuracy of the approach.**

***Index Terms*— Hard sample triplet loss, local image masking, remote sensing image–text retrieval (RSITR), semantic alignment, text-guided reconstruction.**

I. Introduction

REMOTE sensing image–text retrieval (RSITR) is a cross-modal task to retrieve images that match text descriptions or text descriptions that match image queries [1], [2], [3]. This task has widespread applications in land-use monitoring, disaster assessment, and environmental protection [4], [5]. Remote sensing images contain rich geographical information and complex scene features. In contrast, text descriptions provide concise summaries of image content. Effectively combining information from two different modalities to achieve accurate semantic alignment remains a key challenge in RSITR [6], [7], [8].

Recent advancements in deep learning have led to numerous studies dedicated to solving the image–text retrieval task [9], [10]. These methods are typically divided into two key categories: global feature alignment methods and local cross-modal matching methods. Global feature alignment methods address semantic alignment by encoding images and texts into global feature vectors and minimizing their cross-modal distance in the feature space [11], [12]. Zhang et al. [12] proposed an adaptive alignment strategy using curriculum learning to align image–text pairs progressively from easy to difficult. Recent studies increasingly apply vision–language pretrained models [13], [14], [15] to RSITR. Gao et al. [15] improved alignment between images and texts in open-vocabulary settings using carefully designed text prompts, significantly boosting downstream tasks such as image classification, retrieval, and generation. Global feature alignment methods often overlook local details in remote sensing images, failing to capture fine-grained semantics. Local cross-modal matching methods have introduced regional feature extraction and region-word alignment strategies [16], [17]. Yuan et al. [17] integrated a multiscale visual self-attention module to capture prominent features from remote sensing images, which are subsequently employed to enhance the text representations. Recent methods have sought to combine the integrity of global features with the details of local features [18], [19]. Yuan et al. [6] employed local information to refine global features and utilized global information to enhance local details. Wang et al. [18] proposed a multiscale interactive transformer to extract image and text features at different scales. Despite these advances, achieving accurate cross-modal semantic alignment in RSITR task remains a major challenge. In particular, effectively combining local details and global semantics for image–text cross-modal alignment requires further exploration.

In response to the challenges faced in RSITR, this article proposes a context-aware local–global semantic alignment (CLGSA) method. CLGSA aims to align cross-modal semantics at both local and global levels, addressing the need for precise semantic matching between images and textual descriptions. To achieve this, a local key feature alignment (LKFA) module is designed, which introduces a local image masking and reconstruction task. Here, textual context

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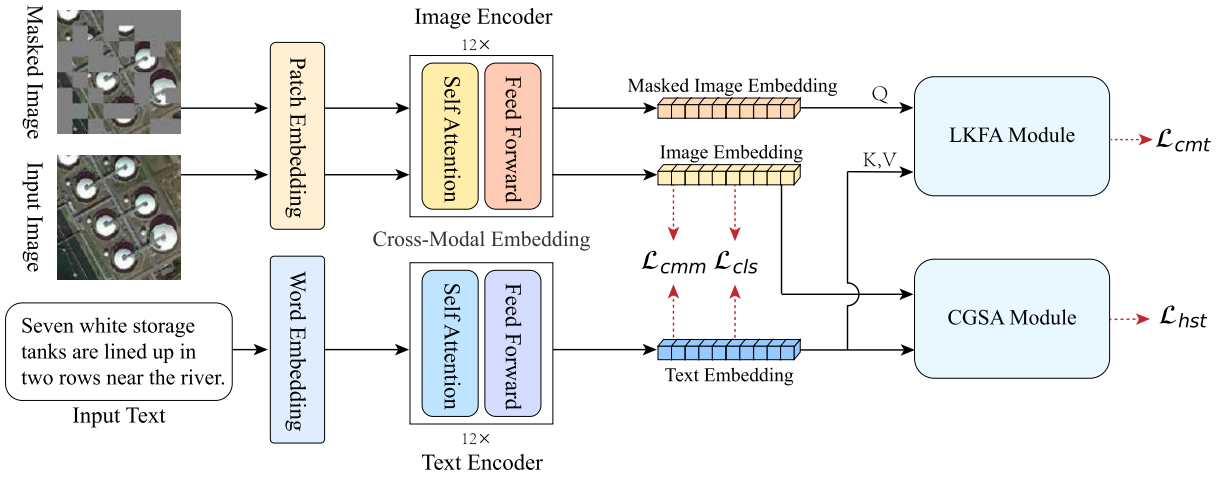


Fig. 1. Illustration of the proposed CLGSA method. CLGSA introduces a context-aware semantic alignment mechanism for both local and global levels. The LKFA module promotes local alignment, and the CGSA module optimizes the global alignment.

information is used to dynamically adjust the local features of the image, thus enhancing the accuracy of local semantic correspondence. This module focuses on aligning specific image regions with corresponding textual terms, improving the model's ability to handle region-specific queries effectively. Simultaneously, a cross-sample global semantic alignment (CGSA) module is constructed to address the global alignment. This module leverages the contextual semantic relationships between samples to guide the model toward a better understanding of the global semantics. This dual-level alignment process, integrating local and global semantic understanding, ensures a more robust and accurate image–text retrieval mechanism.

The general framework of the proposed CLGSA method is depicted in Fig. 1. The proposed method extracts image and text features based on the contrastive language-image pretraining (CLIP) model. First, the input image and text are passed into the image and text encoders in the cross-modal embedding module. Features are extracted under the constraints of intramodal classification loss and cross-modal matching loss to achieve cross-modal matching. Next, the LKFA module masks the local area of the image. It guides the reconstruction of the masked area using text context information. This process is optimized by the contextual masked reconstruction loss to refine the cross-modal local feature alignment. Then, the CGSA module utilizes the cross-modal global contextual relationship between samples. It directs the model to capture the feature patterns of hard-to-differentiate samples through the hard sample triplet loss. The model adjusts the distance in the feature space to enhance global semantic consistency. Finally, the complete network is optimized through an end-to-end training approach under the joint constraints of intramodal classification loss, cross-modal matching loss, contextual masked reconstruction loss, and hard sample triplet loss. During testing, the final retrieval result is generated by calculating the similarity score between the image and text features generated from the encoders.

Comprehensive experiments have been carried out to validate the efficacy of the introduced CLGSA method on two publicly available RSITR datasets, i.e., RSICD [20] and RSITMD [17]. The contributions can be summarized into three points.

1) A novel CLGSA method is proposed, which considers the feature alignment of local fine-grained content of remote sensing images and texts, and also the learning of global semantic matching across samples.
2) An LKFA module is proposed to enhance the fine-grained local feature alignment between remote sensing images and their corresponding texts.
3) A CGSA module is proposed to mine the contextual relationship between samples, which improves the global semantic consistency between remote sensing images and texts.

The structure of this article is as follows. Section II provides a review of related studies. Section III presents the technical details of the proposed method in detail. Section IV shows the experimental design and result analysis. Finally, Section V summarizes the research in this article.

II. Related Work

A. Remote Sensing Image–Text Retrieval

The RSITR task involves identifying the image that most closely aligns with a given text description from a large collection of remote sensing images, or conversely, finding the text that best corresponds to a specific image within a set of textual descriptions [5], [9], [21]. How to map the features of these two modalities to the same semantic space and ensure semantic alignment between them is the core issue in RSITR [6], [22]. Based on different ways of image feature extraction, current RSITR methods can generally be classified into two main groups: deep convolutional neural network (DCNN)-based methods and Transformer-based methods.

DCNN-based methods use convolutional neural networks and recurrent neural networks to extract features from remote sensing images and texts, respectively, and fuse them through different strategies to achieve cross-modal retrieval [6], [7], [23]. Ji et al. [7] proposed a knowledge-assisted momentum

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contrastive learning method to construct a dynamic dictionary containing diverse negative samples, promoting the model to learn discriminative features. Zheng et al. [24] proposed a scale-semantic joint decoupling network that enhances cross-modal retrieval performance. This is achieved through a bidirectional scale decoupling module, a label-supervised semantic decoupling module, and a semantic-guided triple loss. Chen et al. [25] proposed a symmetric multilevel guided network, which introduces bidirectional guidance between modalities to enhance feature learning. Wu et al. [26] introduced a pseudo-region-based spatial–channel attention transformer. Their method improves local object representation and semantic understanding in remote sensing images by using a pseudo-region generation module and a spatial–channel attention mechanism, thus enhancing image–text retrieval performance.

Transformer-based methods process images and texts through two independent Transformer branches, respectively, and then align features through a shared space [23], [27], [28]. Rahhal et al. [29] introduced several architectural models that leverage vision and language transformers to address cross-modal retrieval tasks. Building on this, they introduced a multilingual framework that supports queries in multiple languages (e.g., English, Arabic, French, and Italian) by jointly training language and vision encoders [30]. Li et al. [31] developed an RSITR model based on fusion contrastive learning. This model extracts single-modal features using a self-attention module and improves retrieval performance by incorporating a cross-modal fusion module and contrastive loss function. Liu et al. [32] introduced RemoteCLIP, the first visual-language foundation model for remote sensing. Through data augmentation and the inclusion of drone imagery, this model learns discriminative visual features and text embeddings, significantly enhancing performance in tasks such as image–text retrieval, remote sensing image classification, and target counting. Ma et al. [2] proposed the direction-oriented visual-semantic embedding (DOVE) model. It enhances the alignment of visual and text embeddings in the latent semantic space through a region-oriented attention module and a lightweight text gene mining assistant. Chen et al. [33] proposed a multiscale salient image-guided text alignment method, which adopts multiscale fusion, saliency information learning modules, and an image-guided text alignment mechanism. Although many advances have been made in improving the semantic consistency of images and texts, existing methods still have difficulty in handling subtle differences between images and texts, especially when the descriptions are highly similar or ambiguous. In this case, local semantic alignment of remote sensing images and texts is important.

B. Vision–Language Pretraining

Vision–language pretraining (VLP) aims to jointly model visual and linguistic information [13], [34], [35]. VLP can simultaneously learn the semantic relationship between images and texts, performing various vision–language tasks such as image description generation [36], image–text matching [37], visual question answering [38], and image–text retrieval [2]. Li et al. [39] designed a vision-and-language BERT (ViLBERT) based on a dual-stream structure, using bidirectional encoder representations from transformers (BERT) [40] for text processing and convolutional neural networks for image processing. By applying a multilayer self-attention mechanism, ViLBERT achieves joint modeling of both modalities, enhancing cross-modal understanding. Similarly, Tan and Bansal [41] proposed to learn cross-modality encoder representations from transformers, utilizing a shared Transformer architecture to process both visual and language information simultaneously. Radford et al. [42] introduced a CLIP method to learn a shared representation of images and texts. The model undergoes unsupervised pretraining on a large number of image–text pairs to learn the alignment between images and texts. Chen et al. [43] proposed a universal image–text representation learning method, which learns joint image–text embeddings through large-scale pretraining. Li et al. [44] introduced an object-semantics aligned pretraining method for vision–language tasks, which simplifies the alignment process by using object labels in images as anchors to learn image–text semantic alignment.

Despite the success of visual language pretraining models in natural image processing, their application in remote sensing remains in the exploratory stage [45]. Remote sensing images contain complex geographic information and multidimensional details, such as various object types, spectral data, and geographical context. As a result, remote sensing image–text tasks present greater challenges [27], [33], [46]. Chen et al. [46] proposed a cue-driven multiassociation feature matching network based on CLIP, improving the fine alignment and cross-modal association between remote sensing images and texts. Wang et al. [27] introduced a context and uncertainty-aware prompt method for cross-modal RSITR. The CLGSA method presented in this article also builds on the visual language pretraining model and additionally integrates a context-aware semantic alignment mechanism. CLGSA optimizes the semantic alignment between remote sensing images and texts at two levels: local and global. This dual optimization enhances the performance of cross-modal retrieval.

III. Proposed Method

The general framework of the proposed CLGSA method is illustrated in Fig. 1. The network is composed of three primary components: the cross-modal embedding module, the LKFA module, and the CGSA module. The input image and text are processed by the image and text encoders in the cross-modal embedding module, respectively (Section III-A). Local feature alignment is further optimized in the LKFA module (Section III-B). Global semantic consistency is enhanced by the CGSA module (Section III-C). During training, the whole network is trained end-to-end with all losses. During testing, mere image and text encoders are adopted to extract features for cross-modal similarity ranking.

A. Cross-Modal Embedding

Let the remote sensing image set be $\mathcal{V} = \{I_i\}_{i=1}^{N_v}$ and the corresponding text set be $\mathcal{T} = \{T_i\}_{i=1}^{N_t}$, with N_v and N_t representing the number of images and texts, respectively. The

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image–text pair set in RSITR is given by $\mathcal{S} = \{(I_i, T_i), y_i\}_{i=1}^{N}$, where each image I_i is paired with a text T_i, $y_i \in \mathcal{Y} = \{1, \ldots, N_v\}$ indicates the image label, and N is the number of cross-modal pairs. Inspired by some successful works on transferring knowledge from CLIP [42] to text–image retrieval tasks [2], [23], [33], [46], the proposed method also directly adopts the full image and text encoders of CLIP to initialize the network parameters, aiming to enhance its potential cross-modal alignment capability.

1) Image Encoder: The input image $I_i \in \mathcal{V}$ is divided into N_* nonoverlapping image patches of fixed size $P \times P$. These image patches are then sent to the image encoder along with the [CLS] token. After encoding, the image feature representation is obtained as $V_i = \{v_{\text{cls}}^i, v_1^i, v_2^i, \ldots, v_{N_*}^i\} \in R^{(N_*+1)d}$, where d is the dimension of the latent feature space, $\{v_j^i\}_{j=1}^{N_*}$ is the feature of the image patches, and v_{cls}^i represents the encoded feature of the [CLS] token. Generally, v_{cls}^i can be regarded as the global feature of the input image.

2) Text Encoder: The input text $T_i \in \mathcal{T}$ is first tokenized using a lower cased byte pair encoding (BPE) method [47] with a vocabulary size of 49 152. To indicate the sequence boundaries, the start of sequence (SOS) and end of sequence (EOS) tokens are added at the beginning and end of the token sequence, respectively. The tokenized sequence is then passed through the text extractor, resulting in the text feature representation $T_i = \{t_{\text{sos}}^i, t_1^i, t_2^i, \ldots, t_{N_o}^i, t_{\text{eos}}^i\} \in R^{(N_o+2)d}$, where N_o indicates the length of the input text and $\{t_j^i\}_{j=1}^{N_o}$ represents the word-level feature extracted from the text, while t_{sos}^i and t_{eos}^i correspond to the features of the SOS and EOS tokens, respectively. Typically, the feature of the end token t_{eos}^i is considered the global representation of the entire text.

3) Basic Loss: To learn semantically consistent features from images and texts, the image and text encoders are optimized under two losses: intramodality classification loss and cross-modality matching loss.

The intramodality classification loss expects that image and text features can correctly predict their classes after passing through the classifier [48]. This loss makes features within the same modality more distinct. Specifically, it minimizes the distance between features belonging to the same class and increases the separation between features from different classes. Given a batch of B image–text pair, the intramodality classification loss can be formulated as follows:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B}\sum_{i=1}^{B}\Big(y_i \log\big(\text{softmax}\big(W_{\text{cls}} v_{\text{cls}}^i\big)\big) + y_i \log\big(\text{softmax}\big(W_{\text{cls}} t_{\text{eos}}^i\big)\big)\Big) \tag{1}$$

where W_{cls} represents the parameters of the classifier.

The cross-modal matching loss aligns the feature distributions of images and texts, expecting that features from both modalities can effectively match each other [23], [27]. Construct a batch of B image–text feature pairs $\{(v_{\text{cls}}^i, t_{\text{eos}}^j), y_{i,j}\}_{i,j=1}^{B}$, where $y_{i,j} = 1$ means the image and text pair match, and $y_{i,j} = 0$ means they do not match. For each image–text feature pair, the cosine similarity between their features is first calculated. Following this step, the model can then compute the probability that a given pair is a match as follows:

$$p_{i,j} = \frac{\exp\big(\text{sim}\big(v_{\text{cls}}^i, t_{\text{eos}}^j\big)/\tau\big)}{\sum_{k=1}^{B}\exp\big(\text{sim}\big(v_{\text{cls}}^i, t_{\text{eos}}^k\big)/\tau\big)} \tag{2}$$

where τ is a temperature hyperparameter that regulates the sharpness of the probability distribution. The cross-modal matching loss facilitates feature learning across different modalities. It achieves this by calculating the Kullback–Leibler (KL) divergence of the similarity probability distribution of image–text pair features. Denote $q_{i,j}$ as the true matching probability between the image feature v_i and text feature t_j. The KL loss from image to text is formulated as

$$\mathcal{L}_{i2t} = \text{KL}(p_i \| q_i) = \frac{1}{B}\sum_{i=1}^{B}\sum_{j=1}^{B} p_{i,j} \log\left(\frac{p_{i,j}}{q_{i,j} + \sigma}\right) \tag{3}$$

where σ is a small constant added to prevent numerical instability. For the text-to-image retrieval, the KL loss $\mathcal{L}_{t2i}$ can be calculated similarly. To this end, the cross-modal matching loss capturing the misalignment between image and text features is defined as follows:

$$\mathcal{L}_{\text{cmm}} = \mathcal{L}_{i2t} + \mathcal{L}_{t2i}. \tag{4}$$

By combining $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{cmm}}$, the model promotes consistent feature distribution across modalities. However, further attention needs to be paid to the alignment of local fine-grained features of images and texts, as well as imposing closer constraints between cross-modal global features.

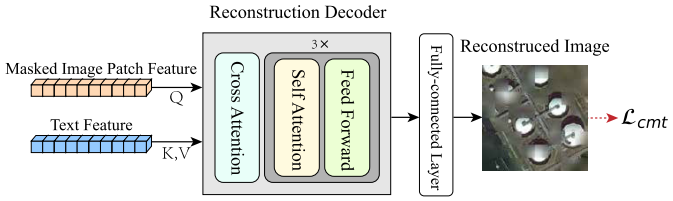


Fig. 2. Illustration of the proposed LKFA module, which uses masked image patch features and text features to reconstruct the masked regions.

B. Local Key Feature Alignment

The proposed method designs an LKFA module as shown in Fig. 2. The core idea is to introduce an additional image local masking and reconstruction task, which uses text context information to guide the reconstruction of masked local areas in remote sensing images. This module encourages interactive learning between the two modalities in local areas and promotes local feature alignment.

1) Feature Encoding: The input image is segmented into a series of N_* equal-sized and distinct patches, ensuring that the image is represented in a grid-like structure of nonoverlapping regions. To encourage the model to better capture the mapping from text to image, a subset of these image patches is randomly masked. The masked patches are then replaced by special noise and mapped into 1-D tokens. Simultaneously, the input text is converted into a series of feature vectors. The masked image tokens and the original text feature vectors are then fed into their respective encoders to extract features. The final hidden states $\{f_n^v\}_{n=1}^{N_p}$ and $\{f_n^t\}_{n=1}^{L}$ of the dual encoders

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will be used to construct the masked image patches, where N_p indicates the number of masked patches and L denotes the word numbers of the input text.

2) Cross-Modal Interactive Decoding: The cross-modal interaction decoder is composed of a multihead cross-attention (MCA) layer and N layers of Transformer [49] block. The output of feature encoders is normalized and serves as the input to the decoder. Specifically, the masked image patch features $\{f_n^v\}_{n=1}^{N_p}$ are utilized as the query $\mathcal{Q}$, and text features $\{f_n^t\}_{n=1}^{L}$ are simultaneously used as the key $\mathcal{K}$ and value $\mathcal{V}$. The decoder evaluates the correlation scores by comparing the query with the key to identify associations between image and text features as follows:

$$\mathrm{MCA}(\mathcal{Q},\mathcal{K},\mathcal{V}) = \mathrm{softmax}\left(\frac{\mathcal{Q}\mathcal{K}^T}{\sqrt{d}}\right)\mathcal{V} \tag{5}$$

where $\mathrm{MCA}(\cdot)$ denotes the multihead cross-attention, and d refers the dimension of the masked tokens. These scores are used to weight the value, generating an image feature vector enriched with text-related information

$$\{z_n\}_{n=1}^{N_p} = \mathrm{Transformer}(\mathrm{MCA}(\mathcal{Q},\mathcal{K},\mathcal{V})) \tag{6}$$

where $\{z_n\}_{n=1}^{N_p}$ represents the contextual embeddings generated from the combined text and masked image features. These embeddings are then passed through a single-layer fully connected network. The network processes the embeddings to produce predicted image patches. Let x_m^n and $\hat{x}_m^n$ represent the predicted value and true value of the mth pixel of the nth image patch, respectively. For the ith input image–text pair, the contextual masked reconstruction loss is represented as

$$\mathcal{L}_{\mathrm{cmr}}^i = \frac{1}{N_p}\sum_{n=1}^{N_p}\sum_{m=1}^{P^2}\left(x_m^n - \hat{x}_m^n\right)^2. \tag{7}$$

Given a collection containing B image–text pairs, the contextual masked reconstruction loss is calculated as

$$\mathcal{L}_{\mathrm{cmr}} = \frac{1}{B}\sum_{i=1}^{B}\mathcal{L}_{\mathrm{cmr}}^i. \tag{8}$$

C. Cross-Sample Global Semantic Alignment

RSITR requires accurate modeling of semantic associations between features from different modalities. During retrieval, remote sensing image features with similar text descriptions or text features describing similar remote sensing scenes should be effectively associated. However, most existing methods overlook potential semantic relationships between cross-modal samples within a batch, limiting retrieval performance. To address this issue, the proposed method designs a CGSA module, which uses the hard sample triplet loss [50] to constrain the sample relationship within the batch and improve the retrieval accuracy and robustness. The core idea is to construct a triplet structure consisting of an anchor, a hard positive, and a hard negative sample to optimize the cross-modal feature representation. The selection process of hard sample triplets is illustrated in Fig. 3.

The choice of hard positive and hard negative samples is determined by computing the cosine similarity between the cross-modal features. For a given image feature v_i, the hard positive sample is defined as the text feature with the lowest cosine similarity among all positive pairs within the batch. The selection process can be expressed as

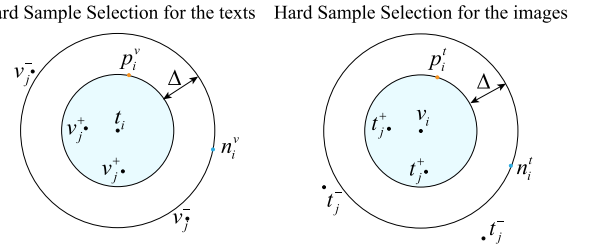


Fig. 3. Selection process of hard sample triplets in the CGSA module, where the hard positive sample is the farthest positive, and the hard negative sample is the closest negative to the anchor feature.

$$p_i^t = \arg\min_{\{t_j | y_{i,j}=1\}} \mathrm{sim}(v_i, t_j) \tag{9}$$

where p_i^t denotes the hard positive text feature for the image feature v_i, $\mathrm{sim}(\cdot,\cdot)$ represents the cosine similarity between two inputs, and $y_{i,j}=1$ means v_i and t_j form a positive pair. Similarly, the process of selecting the hard negative sample entails identifying the text feature most similar to the image feature v_i among all negative pairs within the batch. The selection process is formulated as

$$n_i^t = \arg\max_{\{t_j | y_{i,j}=0\}} \mathrm{sim}(v_i, t_j) \tag{10}$$

where n_i^t denotes the hard negative text feature for the image feature v_i, and $y_{i,j}=0$ means v_i and t_j form a negative pair. Denote the hard positive and negative image features for a text feature anchor t_i as p_i^v and n_i^v, respectively. The hard sample triplet loss is defined as

$$\begin{aligned}\mathcal{L}_{\mathrm{hst}} = &\frac{1}{B}\sum_{i=1}^{B}\left[\Delta + \mathrm{sim}(v_i, n_i^t) - \mathrm{sim}(v_i, p_i^t)\right]_+ \\ &+ \frac{1}{B}\sum_{i=1}^{B}\left[\Delta + \mathrm{sim}(t_i, n_i^v) - \mathrm{sim}(t_i, p_i^v)\right]_+\end{aligned} \tag{11}$$

where Δ is a margin that defines the minimum allowable distance between positive and negative pairs. The hard sample triplet loss is designed to optimize the feature distance between cross-modal pairs by drawing the anchor closer to the most difficult positive sample while simultaneously distancing it from the most challenging negative sample. This loss encourages the model to focus on challenging samples, enhancing both its robustness and generalization ability in cross-modal retrieval tasks.

Overall Loss Function: The proposed CLGSA method is trained in an end-to-end manner, with the overall loss function designed as

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{cmm}} + \mathcal{L}_{\mathrm{cmr}} + \mathcal{L}_{\mathrm{hst}}. \tag{12}$$

The intramodal classification loss $\mathcal{L}_{\mathrm{cls}}$ promotes the class discrimination of intramodal features; the cross-modal matching loss $\mathcal{L}_{\mathrm{cmm}}$ promotes the global semantic alignment of images and texts in a shared semantic space; the contextual masked reconstruction loss $\mathcal{L}_{\mathrm{cmr}}$ helps the model capture the

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fine-grained connection between images and texts in local areas; the hard sample triplet loss $\mathcal{L}_{\text{hst}}$ enhances the model's stability and precision by strengthening the learning of hard samples. Through the incorporation of various loss functions, the proposed method can effectively improve the effect of RSITR. In the experiments, all loss terms have the same magnitude. To ensure that each loss term has a balanced impact on model training, each loss term in (12) is given a coefficient value of 1.

IV. Experiments

A. Datasets and Metrics

Following previous methods [6], [9], [51], the proposed method is thoroughly assessed using two widely recognized standard RSITR datasets. The experimental tasks include text-to-image (T2I) and image-to-text (I2T) tasks. In the T2I task, given a text query, the model needs to retrieve the image that best matches the query text from the image library. In the I2T task, given an image query, the model needs to retrieve the text that best matches the query image from the text library.

RSICD dataset [20] consists of 10 921 high-resolution remote sensing images sourced from Google Earth. Each image has a fixed size of 224 × 224 pixels though the resolution varies across images. Each image is paired with five manually annotated descriptive texts. However, only 724 images have five distinct captions, while the rest contain repeated captions. As a result, the dataset contains 24 333 sentences. This high repetition rate and the variation in image resolution present additional challenges for the dataset. Following previous methods [6], [17], the dataset is divided into 7862 training images, 1966 validation images, and 1093 test images.

RSTMD dataset [17] contains 4743 remote sensing images, each with a size of 256 × 256 pixels. These images are sourced from the RSICD dataset and Google Earth. For each image, there are five corresponding descriptive texts. The annotation process of the RSTMD dataset emphasizes fine-grained relationships between objects. As a result, the sentences in this dataset provide more detailed and specific information compared to the RSICD dataset. The dataset is divided into 3435 training images, 856 validation images, and 452 test images.

Metrics: Following previous methods [1], [3], [27], $R@k$ and mR are used to evaluate the performance of retrieval algorithms. $R@k$ is defined as the ratio of the number of correct targets found in the top k retrieval results to the total number of test samples. A higher $R@k$ value indicates that the model has a higher hit rate when retrieving targets. This article reports $R@1$, $R@5$, and $R@10$ for both T2I and I2T retrievals. The mR metric is the average of the six $R@K$ indicators mentioned above. This metric provides a comprehensive reflection of the model's retrieval performance at different ranking depths. It avoids focusing solely on results from a few fixed positions and thus offers a more thorough evaluation of the model's retrieval performance. A higher value of the metric mR reflects that the model performs well across multiple retrieval depths.

B. Experimental Configurations

The proposed network consists of two encoders along with a contextual masked reconstruction decoder. For initialization, the image and text encoders leverage pretrained parameters from the CLIP model, ensuring that both modalities benefit from prior knowledge and established feature extraction capabilities. In contrast, the parameters of the decoder within LKFA module are initialized randomly. Each layer of the decoder is configured with a hidden feature size of 512 and 8 attention heads. The input images are resized to 256 × 256 pixels, with each patch having a size of 16 × 16. During the training process, data augmentation techniques including random horizontal flipping, cropping with padding, and erasing are employed. The maximum length for the textual token sequence is set to 77. The Adam optimizer is used to optimize the network training over 60 epochs. The initial learning rate is set as 1×10^{-5}. A cosine learning rate decay is implemented throughout the training. The learning rate gradually increases from 1×10^{-6} to 1×10^{-5} during the first five warm-up epochs. For the parameters initialized randomly, the starting learning rate is set to 5×10^{-5}. Additionally, the temperature parameter τ is fixed at 0.02, the margin Δ is set to 0.05, and the small constant σ is set to 1×10^{-8}. The proposed method has been implemented using PyTorch, leveraging a single NVIDIA GeForce RTX 3090 GPU.

C. Ablation Study

To ascertain the precise contributions of the four loss functions ($\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{cmm}}$, $\mathcal{L}_{\text{cmr}}$, and $\mathcal{L}_{\text{hat}}$) integrated within the proposed method, a comprehensive series of ablation experiments were conducted, as delineated in Table I. The comparative analyses are presented as follows.

1) $\mathcal{L}_{cls}$*:* The first experimental configuration in Table I corresponds to the model trained exclusively with the intramodal classification loss $\mathcal{L}_{\text{cls}}$, serving as the baseline model. The baseline model achieves 4.26% $R@1$ score on RSICD dataset and 10.40% on RSITMD in the T2I task. These results indicate that while the baseline model effectively ensures the fundamental discriminability of features within each modality, its performance in cross-modal retrieval tasks remains suboptimal. This limitation can be attributed to the absence of mechanisms for cross-modal alignment and fine-grained detailed capture, which are essential for enhancing retrieval effectiveness.
2) $\mathcal{L}_{cls} + \mathcal{L}_{cmm}$*:* The second experimental configuration introduces the cross-modal matching loss $\mathcal{L}_{\text{cmm}}$ to the baseline model. This addition significantly enhances the model performance. Specifically, the mR metric improves by +7.36% on RSICD and +12.48% on RSITMD, respectively. These experimental results indicate that the cross-modal matching loss plays a crucial role in effectively aligning the image and text features within a unified space, leading to improved retrieval accuracy.
3) $\mathcal{L}_{cls} + \mathcal{L}_{cmr}$*:* The third experimental configuration incorporates contextual masked reconstruction loss into the

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TABLE I
ABLATION EXPERIMENTAL RESULTS ON RSICD AND RSITMD DATASETS

	Losses				RSICD dataset							RSITMD dataset						
					T2I			I2T			mR	T2I			I2T			mR
	$\mathcal{L}_{cls}$	$\mathcal{L}_{cmm}$	$\mathcal{L}_{cmr}$	$\mathcal{L}_{hst}$	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
(1)	✓				4.26	15.76	26.57	5.30	15.19	25.80	15.48	10.40	25.66	36.95	8.32	28.27	47.43	26.17
(2)	✓	✓			8.62	25.38	39.98	7.77	21.50	33.76	22.84	18.37	42.70	50.26	17.65	44.37	58.54	38.65
(3)	✓		✓		7.96	23.42	35.42	6.92	18.26	29.54	20.25	17.48	35.40	46.90	15.49	43.89	65.09	37.37
(4)	✓	✓	✓		12.85	35.94	43.07	10.72	32.30	45.57	30.07	20.35	46.42	56.64	19.74	44.69	52.92	40.13
(5)	✓	✓	✓	✓	**14.38**	**36.73**	**45.35**	**11.75**	**34.07**	**50.12**	**32.07**	**26.77**	**48.23**	**60.39**	**22.52**	**52.39**	**69.46**	**46.63**

baseline model. This modification results in performance improvements compared to the baseline model. $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{cmr}}$ increases mR metric by +4.77% on RSICD and +11.2% on RSITMD. However, this improvement is smaller than that achieved by $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{cmm}}$. The experimental results demonstrate that contextual masked reconstruction loss helps to capture detailed features in images and texts, enhancing retrieval precision and recall. However, enforcing cross-modal consistency of global features has a more pronounced impact on the RSITR performance compared to fine-grained detailed alignment.

4) $\mathcal{L}_{cls}+\mathcal{L}_{cmm}+\mathcal{L}_{cmr}$: The fourth experimental configuration builds on $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{cmm}}$ by incorporating the contextual masked reconstruction loss $\mathcal{L}_{\text{cmr}}$. This addition further improves model performance. Specifically, compared with $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{cmm}}$, this configuration increases $R@1$ by +4.23% on RSICD and +1.98% on RSITMD. Also, mR improves by +7.23% and 1.48% on the two datasets, respectively. These improvements result from the ability of $\mathcal{L}_{\text{cmr}}$ to capture detailed connections between cross-modal local regions. This capability helps the model to accurately understand and match key information in images and texts, complementing global feature cross-modal matching.

5) $\mathcal{L}_{cls}+\mathcal{L}_{cmm}+\mathcal{L}_{cmr}+\mathcal{L}_{hst}$: The fifth experimental configuration incorporates the hard sample triplet loss based on $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{cmm}} + \mathcal{L}_{\text{cmr}}$. For T2I task, the model achieves $R@1$ scores of 14.38% on RSICD and 26.77% on RSITMD. The mR scores are 32.07% and 46.63% on these two datasets, respectively. These results demonstrate that the hard sample triplet loss promotes the model to achieve generalization ability and retrieval performance. Remote sensing images often feature complex scenes with subtle visual differences. As a result, some image–text pairs can be highly similar or difficult to distinguish. The hard sample triplet loss addresses this challenge by encouraging the model to focus on these challenging examples. It promotes the model to understand cross-modal complex relationships. Consequently, it enhances both the accuracy and robustness of retrieval performance in the RSITR task.

The above ablation study results demonstrate the effectiveness of the proposed CLGSA method, which integrates four different loss functions. Each individual loss function plays a vital role in improving the performance of the model in a specific aspect. The change curve of each loss term during training is shown in Fig. 4. As shown in the figure, the four loss terms show similar magnitudes and all show a clear downward trend as the training progresses, indicating that the model has been effectively optimized and converged. In addition, additional experiments are conducted to study the impact of changing the coefficient of the larger loss term $\mathcal{L}_{\text{cmm}}$. The results are shown in Table II, indicating that adjusting the weight of $\mathcal{L}_{\text{cmm}}$ does not lead to significant performance differences. Therefore, the proposed method sets the coefficient of each loss term to 1 to achieve stable optimization and effective convergence.

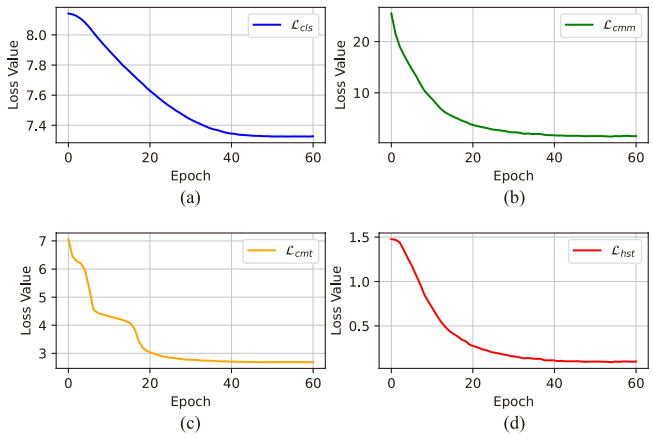


Fig. 4. Change curve of each loss with epoch during training on the RSITMD dataset. (a) $\mathcal{L}_{\text{cls}}$. (b) $\mathcal{L}_{\text{cmm}}$. (c) $\mathcal{L}_{\text{cmr}}$. (d) $\mathcal{L}_{\text{hst}}$.

TABLE II
EVALUATION OF DIFFERENT $\mathcal{L}_{\text{CMM}}$ COEFFICIENTS

Coefficient	mR	
	RSICD dataset	RSITMD dataset
1.0	32.07	46.63
0.9	31.82	46.32
0.7	31.58	46.01
0.5	31.32	45.78
0.3	30.95	45.61
0.1	30.73	45.48

D. Parameter Analysis

These experiments aim to evaluate the effect of different parameters on the method's performance. Key settings are

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TABLE III
EVALUATION OF DIFFERENT MASK RATIOS

Mask Ratio	mR	
	RSICD dataset	RSITMD dataset
0.5	28.27	43.99
0.6	30.38	45.19
0.7	32.07	46.63
0.8	29.67	45.06
0.9	21.98	38.83

TABLE IV
EVALUATION OF DIFFERENT IMAGE SIZES

Image Size	mR	
	RSICD dataset	RSITMD dataset
64×64	19.46	24.55
128×128	26.33	37.04
256×256	32.07	46.63
512×512	29.76	42.07

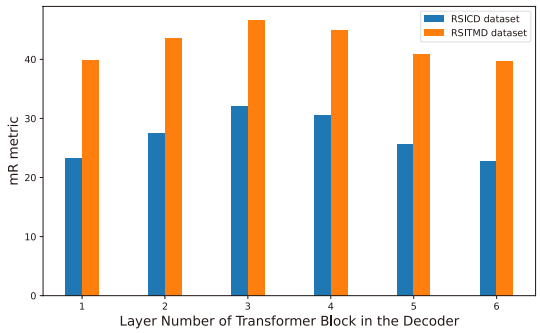


Fig. 5. Experimental analysis of the layer number of transformer blocks in the decoder on RSICD and RSITMD datasets.

adjusted to better understand their influence on the overall results.

1) Mask Ratio: This section compares the model performance with varying mask ratios. Table III presents the mR values of the model on two datasets under different mask ratio settings. As shown, the performance improves as the mask ratio increases from 0.5 to 0.7, followed by a decline when the mask ratio reaches 0.9. This phenomenon can be attributed to the increased difficulty introduced by the higher mask ratio. When the mask ratio is set to 0.9, a larger portion of the image is masked, leading to more significant disruptions in the visual content. As a result, the model has to rely more on the surrounding unmasked regions and the textual context to reconstruct the masked areas. This increases the challenge of aligning image and text features, which may explain the performance drop observed at this higher mask ratio. The best performance is achieved at a mask ratio of 0.7, with mR values of 32.07% on RSICD and 46.63% on RSITMD. These results indicate that an appropriate mask ratio boosts the model's proficiency to capture and align local features in images and texts.

2) Image Size: This section further investigates the impact of input image size on the performance of the proposed method. As shown in Table IV, image sizes of 64 × 64, 128 × 128, 256 × 256, and 512 × 512 are evaluated. The results indicate that larger image sizes generally improve retrieval performance. However, performance is not optimal at 512 × 512. This reduction can be attributed to several factors. First, while high-resolution images capture finer details, they also introduce more noise and irrelevant features, which can hinder focus on the most important semantic elements. Furthermore, the larger image size increases the amount of information the model needs to process, which raises the computational complexity during training due to the larger number of pixels. This results in higher memory and processing requirements, which can impact the model training efficiency and its ability to converge. These challenges highlight the fact that simply increasing image resolution does not always yield better performance. The model achieves the highest mR on two datasets with a size of 256 × 256. This suggests that 256 × 256 strikes a balance between capturing global semantics and fine-grained features.

3) Layer Number of the Transformer Block in the Decoder: This section compares the model effects using different layers of Transformer block, as shown in Fig. 5. The results present a noticeable improvement in performance with the increase in the number of layers. However, when the number of layers exceeds 3, the performance gain becomes less significant. This suggests that the model has already learned the essential features, and additional layers contribute less to feature extraction. Moreover, too many layers may lead to overfitting. When the number of layers is 3, the model achieves the best mR on two datasets. This demonstrates that the three-layer Transformer block strikes a suitable compromise between feature extraction and model complexity, capturing deep semantic information while avoiding overfitting.

E. Comparison With the State-of-the-Art Methods

In this section, the proposed method is compared with several state-of-the-art methods on two benchmark datasets. The compared methods are categorized into two groups: traditional deep learning-based methods and the latest CLIP-based methods. On both two datasets, the proposed method outperforms most compared methods in both $R@k$ and mR metric, demonstrating clear advantages.

The Traditional Deep Learning-Based Methods: Some methods focus on extracting local information from remote sensing images. For instance, SCAT-PRG [26] enhances the extraction of local features in salient objects using a spatial–channel attention transformer. SWAN [51] improves scene awareness and reduces spatial semantic confusion by establishing correlations across features at different granularities. These methods emphasize key regions and features, improving retrieval accuracy. Other methods aim to construct a robust and precise common semantic space for images and texts. SIRS [9] employs a multitask joint learning framework, combining semantic segmentation and image–text retrieval. GaLR [6] integrates features at different levels through a dynamic fusion module. KAMCL [7] enhances the representation of image–text pairs by expanding negative sample pairs using momentum contrastive learning. SMLGN [25]

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TABLE V
COMPARISON OF THE PROPOSED METHOD WITH THE STATE-OF-THE-ART METHODS ON THE RSICD DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	RSICD dataset						
	T2I			I2T			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
SIRS[9]	5.76	20.50	33.78	7.34	20.00	30.78	19.69
GaLR[6]	6.59	19.85	31.04	4.69	19.48	32.13	18.96
SWAN[51]	7.41	20.13	30.86	5.56	22.26	37.41	20.61
SMLGN[25]	8.87	25.53	37.24	7.85	27.14	42.58	24.87
SCAT-PRG[26]	10.88	26.32	39.66	8.06	25.43	40.31	25.11
KAMCL[7]	11.99	27.17	38.33	7.78	25.23	40.02	25.08
DOVE[2]	6.04	23.95	40.35	8.66	22.35	34.95	22.72
MSITA[33]	6.13	21.98	35.39	8.67	22.71	33.91	21.47
CLIP-Ada[15]	7.11	19.48	31.01	7.67	24.87	39.73	21.65
CDMAN[46]	7.13	25.38	40.35	10.23	24.94	36.83	24.14
VGSGN[3]	8.33	21.87	32.57	6.53	23.13	36.85	21.55
CUP[27]	9.86	31.60	47.15	12.05	32.66	48.70	30.34
MSA-CLIP[23]	10.16	25.71	36.96	7.87	25.67	41.85	24.70
PERSVL[1]	10.70	27.08	38.43	10.14	30.43	45.58	27.06
PE-RSITR[52]	14.13	31.51	44.78	11.63	33.92	**50.73**	31.12
CLGSA	**14.38**	**36.73**	**45.35**	**11.75**	**34.07**	50.12	**32.07**

TABLE VI
COMPARISON OF THE PROPOSED METHOD WITH THE STATE-OF-THE-ART METHODS ON RSITMD DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	RSITMD dataset						
	T2I			I2T			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
SIRS[9]	11.98	41.29	60.15	15.25	35.01	48.73	35.40
SWAN[51]	13.35	32.15	46.90	11.24	40.40	60.60	34.11
GaLR[6]	14.82	31.64	42.48	11.15	36.68	51.68	31.41
SCAT-PRG[26]	16.50	33.25	48.49	12.55	41.86	61.07	35.62
KAMCL[7]	16.81	34.96	47.12	14.60	41.86	59.86	35.87
SMLGN[25]	17.26	39.38	51.55	13.19	43.94	60.40	37.62
MSITA[33]	12.15	39.92	57.72	15.22	34.20	47.65	34.48
DOVE[2]	12.20	49.93	66.50	16.81	36.80	49.93	37.73
CLIP-Ada[15]	12.83	28.84	39.05	13.30	40.20	60.06	32.38
CDMAN[46]	13.82	41.96	59.74	16.46	37.74	51.24	36.83
VGSGN[3]	14.16	34.96	50.66	13.23	42.57	63.41	36.50
CUP[27]	17.70	48.40	64.82	21.57	43.70	56.75	42.16
PERSVL[1]	21.68	42.48	57.30	15.44	47.39	65.66	41.66
MSA-CLIP[23]	22.35	42.92	55.75	15.18	47.35	64.73	41.38
PE-RSITR[52]	23.67	44.07	60.36	20.10	50.63	67.97	44.47
CLGSA	**26.77**	**48.23**	**60.39**	**22.52**	**52.39**	**69.46**	**46.63**

introduces intramodal and intermodal guidance through a symmetric multilevel guidance network. By associating images and texts more effectively, these methods aim to improve the model's accuracy and robustness.

The Latest CLIP-Based Methods: Some methods focus on efficiently transferring visual-language pretrained models from the natural image domain to remote sensing tasks. For instance, CLIP-Adap [15] enhances downstream performance by aligning images and texts in an open-vocabulary setting with carefully crafted text prompts. CUP [27] introduces context and uncertainty-aware prompt networks to improve pretrained model performance on remote sensing images. PE-RSITR [52] facilitates parameter-efficient transfer through the design of multimodal adapters and contrastive loss functions. These approaches leverage pretrained knowledge from the natural image domain to enable effective transfer learning for remote sensing applications. Other methods aim to improve feature representation and alignment quality in cross-modal matching. For example, MSITA [33] uses a multiscale salient image-guided text alignment strategy. DOVE [2] explores image–text relationships with a DOVE model. MSA-CLIP [33] enhances cross-modal feature learning through multiscale alignment. CDMAN [46] employs a cue-driven multiassociative feature matching network for fine-grained semantic alignment. VGSGN [3] integrates global saliency guidance with structural information to optimize cross-modal alignment. PERSVL [1] focuses on extracting visual features and enhancing semantic interaction in remote sensing guided by prior experience.

1) Compared Results on RSICD Dataset: The comparison results on RSICD dataset are presented in Table V. Under the $R@k$ metric, the proposed method demonstrates significant advantages. For instance, on the $R@1$ metric, it achieves accuracies of 14.38% in T2I task and 11.75% in I2T task, respectively, outperforming all the comparative methods. Specifically, compared to the deep learning-based method KAMCL [7], the proposed method improves by +2.37% and +3.97% in the two settings. Compared to the CLIP-based method PE-RSITR [52], it improves by +0.25% and +0.12%, respectively. These results indicate that the proposed method can more accurately match query samples at the top of the retrieval results. On the mR metric, the proposed method also performs well. Compared to the deep learning-based method SCAT-PRG [26], the mR score of the proposed method increases by +6.96%. Compared to the CLIP-based method PE-RSITR [52], the mR score improves by +0.95%. These results demonstrate that the proposed method maintains high retrieval accuracy across different ranking depths, consistently returns relevant remote sensing images, and exhibits small fluctuations in ranking depth.

2) Compared Results on RSITMD Dataset: Table VI presents the comparison results on RSITMD dataset. On the RSITMD dataset, the proposed method also shows strong performance. Compared to the deep learning-based method SMLGN [25], the proposed method improves the $R@1$ score by +9.51% in T2I task and +9.33% in I2T task. These results suggest that the proposed method successfully enhances the alignment of image and text features, thus improving retrieval accuracy. Additionally, the proposed method increases the mR score by +9.01%, further demonstrating its significant advantage in improving recall performance. When compared to the CLIP-based method PE-RSITR [52], the proposed method also achieves better performance. Specifically, the proposed method achieves a +2.16% improvement in the mR score. In terms of the $R@1$ score, it improves by +3.10% in T2I task and +2.42% in I2T task. The introduction of contextual

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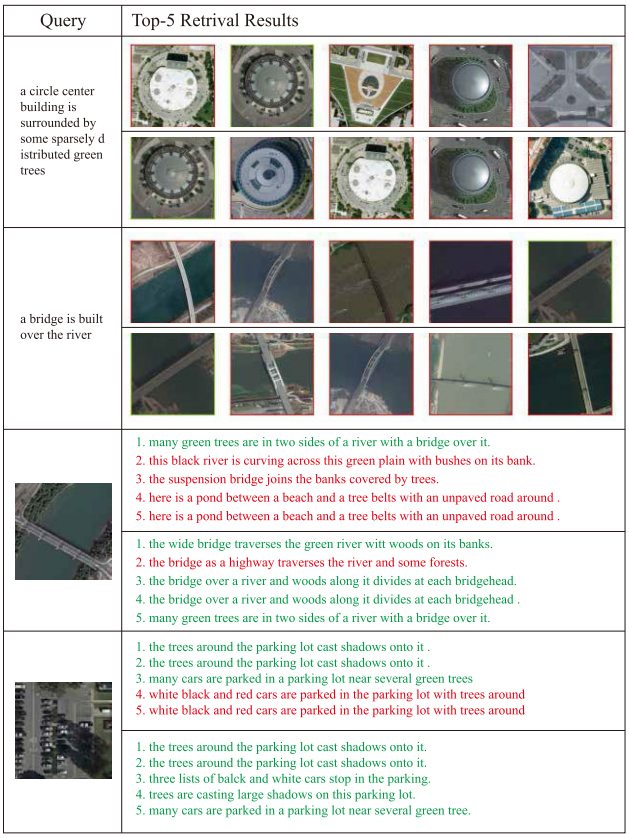


Fig. 6. Examples of retrieval results on the RSICD dataset are presented. For each I2T or T2I query, the first row displays the retrieval results of the baseline model, while the second row shows the results of the proposed method. Matched and mismatched images (or texts) are indicated in green and red, respectively.

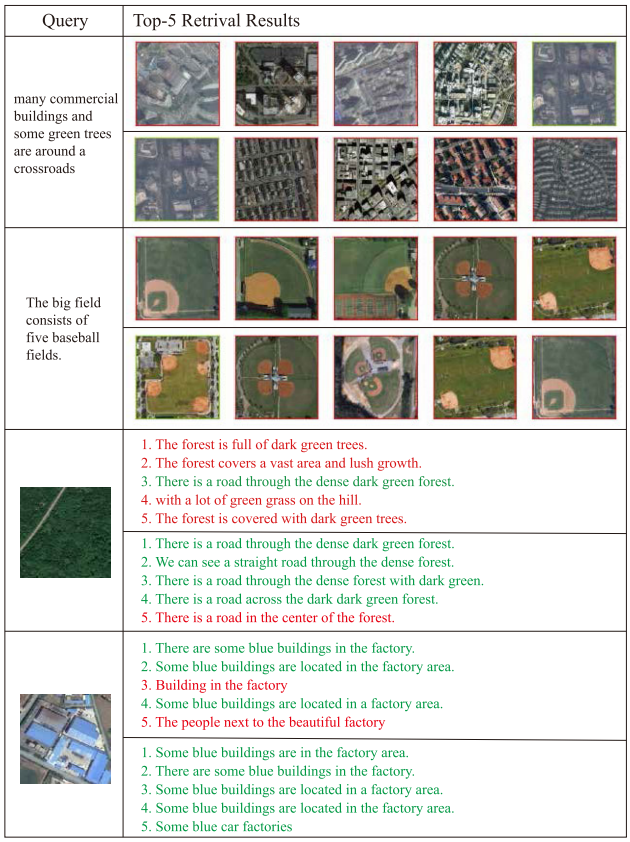


Fig. 7. Examples of retrieval results on the RSITMD dataset are presented. For each I2T or T2I query, the first row displays the retrieval results of the baseline model, while the second row shows the results of the proposed method. Matched and mismatched images (or texts) are indicated in green and red, respectively.

information into images and texts promotes the model to understand fine-grained local features, further improving both retrieval accuracy and recall capabilities. Moreover, the inclusion of the hard sample triplet loss enables the model to learn more effectively from difficult-to-distinguish samples. This reduces errors on boundary samples and improves model stability.

F. Qualitative Experiments

To offer a straightforward and visual demonstration of the proposed method's efficacy, this section presents a qualitative comparison of retrieval results between the baseline model and the proposed CLGSA method.

1) Qualitative Experiments on the RSICD Dataset: The experimental results on the RSICD dataset are presented in Fig. 6. In the T2I task, the proposed CLGSA method retrieves images that closely match the target object based on the text query. For example, given the query "a circle center building is surrounded by some sparsely distributed green trees," all images returned by the proposed method contain the target object "circle center building." In contrast, the fifth image returned by the baseline model does not include the main target. This demonstrates that the proposed CLGSA method exceeds the baseline model in accurately matching the target object. In another example, with the query "a bridge is built over the river," both the baseline model and the proposed method retrieve images featuring a bridge over a river. However, the baseline model fails to rank the correct image first, while the proposed method effectively improves the relevance of image ranking, ensuring the most relevant image is placed at the top. This highlights the proposed method's superior performance in image–text matching accuracy and ranking optimization.

In the I2T task, the proposed method retrieves text descriptions that are highly consistent with the image content. For instance, when the query image shows a bridge over a river, all the text descriptions returned by the proposed method accurately include terms like "bridge" and "river," aligning closely with the image. In contrast, the baseline model returns descriptions containing objects such as "pond," "beach," and "belt," which are similar to the scene but not precise, reflecting its lack of content accuracy. In the second example, the text retrieved by the baseline model incorrectly described the shadows in the image as trees, while the proposed method returned all correct results. This demonstrates that the proposed method achieves stronger semantic consistency and accuracy in the I2T task, effectively avoiding mismatches and providing more accurate text descriptions.

2) Qualitative Experiments on the RSITMD Dataset: The retrieval results on RSITMD dataset are illustrated in Fig. 7. In the T2I task, for the query text "many commercial buildings and some green trees are around a crossroads," all images

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retrieved by the proposed method accurately include key targets such as "commercial buildings" and "crossroads." The correct result is ranked first. In contrast, the baseline model ranks the correct result fifth. For another query, "The big field consists of five baseball fields," both the baseline model and the proposed method retrieve images containing baseball fields. However, none of the baseline results accurately reflect the number of baseball fields. The proposed method ranks the image with five baseball fields first and orders the results such that the number of fields decreases progressively. This demonstrates the significant advantages of the proposed method in matching accuracy and ranking optimization.

In the I2T task, the proposed method retrieves text descriptions closely related to the query image. For example, given an image showing a road cutting through a northwest forest, the proposed method returns descriptions that include keywords like "road" and "forest," which perfectly match the image content. In contrast, the baseline model returns descriptions where the significant target "road," is not mentioned in four out of the five results. In another example, the query image depicts a factory with a blue roof. The proposed method consistently returns descriptions containing keywords such as "blue" and "factory." On the other hand, the baseline model provides two descriptions that are clearly biased and unrelated to the actual content of the image.

Despite its strong performance, the proposed method has areas for improvement. For example, in the first query, the road's direction is incorrectly described as "center" instead of "northwest." Future research should focus on enhancing semantic fine-grained matching, improving the model to understand complex text descriptions and detailed image features. Furthermore, robustness to ambiguous text and image data should be strengthened to reduce mismatches and improve overall performance.

V. Conclusion

This article introduces a novel CLGSA method for RSITR. The method addresses both local and global semantic alignments, which are critical for achieving high-accuracy retrieval in complex remote sensing scenarios. The LKFA module improves the fine-grained alignment by establishing explicit local relationships between local images and specific texts. It employs contextual masked reconstruction to strengthen the image–text matching process. The CGSA module enhances global semantic consistency using a hard sample triplet loss. This strategy enables the model to handle hard-to-distinguish samples effectively. The integration of context-aware local alignment and global consistency provides a robust and accurate solution for RSITR. Future research can further optimize these modules by incorporating advanced multimodal learning techniques or adapting the approach to handle more complex remote sensing data.

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